MIL-B-3753E
30 June 1986
SUPERSEDING
MIL-B-3753D
9 December 1974

MILITARY SPECIFICATION

BUCKETS, CONCRETE, HAND-OPERATED, 3/4- TO 2-CUBIC YARD CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand-operated concrete handling buckets.

1.2 Classification. The bucket shall be one of the following sizes, as specified (see 6.2):

Size 75 - 3/4-cubic yard capacity.
Size 100 - 1-cubic yard capacity.
Size 150 - 1-1/2-cubic yard capacity.
Size 200 - 2-cubic yard capacity.

1.3 Specification part number. The specification part number is a definitive part number which will be formatted to identify each item covered by this specification. The part number will be formatted by selecting from the requirement options in this specification as follows (see 6.2).

Definitive specification part number — M — 3753 — X

M Prefix-An item identified by inch-pound (US) units

Specification number

Size (see 1.3.1

Example: M375B designates a one cubic yard capacity bucket.

1.3.1 Size designator. A one-position numeric field used to show size of bucket (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Capacity designation.

Code		Definition
A	Size 75	3/4 - cubic yard capacity
B	Size 100	1 - cubic yard capacity
C	Size 150	1-1/2 - cubic yard capacity
D	Size 200	2 - cubic yard capacity

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the

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issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 283 - Specification for Low and Intermediate Tensile Strength Carbon Steel Plates of Structural Quality.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.
D14.3 - Earthmoving and Construction Equipment, Specification for Welding.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bucket shall consist essentially of a frame, hopper, discharge gate equipped with a manual control handle or lever, and lifting bail.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the production buckets from the approved first article during

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production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish buckets conforming to this specification.

3.3 Commercial bucket parts and components. Except as modified or required by this specification, the bucket parts and components shall be the contractor's commercial items.

3.4 Material. Material shall be as specified herein. Any material not definitely specified shall be of the quality used by the concrete bucket manufacturer in his commercial product.

3.4.1 Material deterioration prevention and control. The buckets shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the buckets may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bucket may be newly fabricated from recovered materials to the maximum extent practicable, provided the bucket produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bucket.

3.5 Capacity. Rated bucket capacities shall be as specified in 1.2.

3.6 Components.

3.6.1 Hopper. The hopper construction shall assure discharge of the full load. The interior shall be free from projections, indentations, or imperfections that would tend to retard or prevent complete discharge of the contents. The hopper volume shall be not less than 110 percent of the rated capacity. Steel sheet used shall meet or exceed the requirements of ASTM A 283, grade D, and the sheet thickness shall be not less than the following:

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Bucket size	Shell thickness (inches)	US revised standard gage number
75	0.1345	10
100	0.1644	8
150	0.1644	8
200	0.1943	6

3.6.2 Discharge gate. The discharge opening shall be equipped with a quick-opening, positive-closing gate which permits complete discharge of the load. The discharge gate, or any part of the gate mechanism, when closed or in any open position shall not extend below the base of the frame. The gate shall be constructed to hold and transport a capacity load of concrete without leakage. Cement grout leakage shall not exceed 0.5 cubic foot per 5 minute interval after gate closure. There chall be no leakage of concrete 1 minute after closing the gate subsequent to the discharge of a partial load. The opening and closing of the gate shall be controlled by a handle or lever. The handle or lever shall require a force not to exceed 35 pounds to cause complete discharge of the hopper contents. The gate shall be of the self-closing type so that if the handle is released the gate will automatically close and stay closed until the handle is again operated by the operator. The gates shall be of material having strength and wear characteristics suited to the purpose. The control device for the gate shall be located as not to injure or endanger operator while operating the device or upon the opening of the discharge gate.

3.6.3 Lifting bail. The bucket shall be provided with a bail. The bail shall be installed in a manner and in a location that will cause the bucket to balance vertically when the bucket is suspended full or empty. The bail shall be capable of handling the fully loaded bucket under any handling conditions that might be encountered, such as rapid hoisting, swinging and lowering, and sudden braking of the load.

3.7 Safety. All rotating or reciprocating parts that are so located as to be hazardous to operating and servicing personnel, shall be guarded to the extent necessary to eliminate the hazard. Excessive projections of nuts and bolts, sharp edges, points, and rough surfaces shall be eliminated except when required or for normal operation of the bucket.

3.8 Lubrication. Wear between parts shall be minimized by use of bearings, replaceable bushings, or lubrication. Lubrication fittings shall be provided in accessible protected locations where lubrication is required. The bucket shall be lubricated prior to delivery with MIL-G-10924.

3.9 Identification marking. The bucket shall be marked for identification in accordance with MIL-STD-130.

3.10 Treatment and painting. The portion of the bucket normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be green 385 conforming to MIL-C-46168 or MIL-C-53039 as applicable.

3.11 Workmanship. All parts, components, and assemblies of the buckets including castings, forgings, molded parts, stampings, bearings, seals, machined

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surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.11.1 Metal fabrication. Metal used in the fabrication of the buckets shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.11.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welders and welding.

3.11.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding code, Section 5, Qualification.
AWS D14.3 - Earthmoving and Construction Equipment, Specification for Welding.
ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Certification of welder qualification shall be made available for review by the contracting officer or designated representative. Contractors who make only horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS OR ASME codes.

3.11.4.2 Welding. Surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when parts connected by welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bucket to conform to examination and test requirements specified in section 4. Parts to be joined by fillet welds

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shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch. The welding process shall be at the option of the contractor and shall conform to the provisions of AWS D1.1 and D14.3.

3.11.5 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspecition (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bucket(s) shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article bucket(s) shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-1.

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4.4.2 Examination.

4.4.2.1 Individual. Each bucket shall be examined for the defects marked "X" in column 3 of table II. Presence of one or more defects shall be cause for rejection of the bucket.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the defects marked "X" in column 2 of table II. AQL shall be 1.0 percent defective.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table III. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Examination shall be in accordance with table II.

TABLE II. Examination schedule.

First article	Quality conformance: Samples	Quality conformance: Individual	Defect	Requirement paragraph
1	2	3	4	5
			Major	
X	-	X	101. Not size of bucket specified.	1.2
X	X	-	102. Parts or components missing or not as specified.	3.3
X	X	-	103. Materials not as specified.	3.4
X	X	-	104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.4.1
X	X	-	105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.4.2
X	X	-	106. Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.4.3

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance: Samples	Quality conformance: Individual	Defect	Requirement paragraph
1	2	3	4	5
			Major	
X	X	-	107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the bucket.	3.4.4
X	X	-	108. Hopper construction not as specified.	3.6.1
X	X	-	109. Discharge gate missing or not as specified.	3.6.2
X	X	-	110. Lifting bail missing or not as specified.	3.6.3
X	X	-	111. Safety not as specified.	3.7
X	X	-	112. Treatment and painting not as specified.	3.10
X	X	-	113. Paint color not as specified.	3.10
X	X	-	114. Workmanship not as specified.	3.11
X	X	-	115. Metal fabrications not as specified.	3.11.1
X	X	-	116. Bolted and riveted connections not as specified.	3.11.2 and 3.11.3
X	X	-	117. Welders and welding not as specified.	3.11.4.1 and 3.11.4.2
X	X	-	118. Castings not as specified.	3.11.5
			Minor	
X	-	X	201. Bucket not lubricated as specified.	3.8
X	X	-	202. Identification marking not as specified.	3.9

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4.5.2 Tests. Tests shall be in accordance with table III.

TABLE III. Test schedule.

First article	Quality conformance Samples	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	X	Concrete test.	4.5.2.2	3.5, 3.6.1 to 3.6.3
X	X	Grout slurry test.	4.5.2.3	3.5, 3.6.1 to 3.6.3

4.5.2.1 Test conditions. Prior to test, the bucket shall be checked to insure that it has been lubricated in accordance with 3.8. Tests shall be performed without shelter and at the climatic conditions existing at the place of test.

4.5.2.2 Concrete test. The bucket shall be loaded to 110 percent of rated capacity with fresh wet concrete of a consistency of not less than 2-inch and not more than 4-inch slump. The bail of the loaded bucket shall then be secured to the hoisting line of a suitable crane or hoisting device. The loaded bucket shall be raised to a height of 25 feet and lowered by gravity and braked suddenly and completely to a stop with bottom of bucket approximately 5 feet off the ground. The raising, lowering, and braking shall be continued for two additional cycles. Gates shall then be opened to discharge approximately one-half of the concrete and then closed. The gates shall again be operated to discharge the remainder of the concrete. Inability to be loaded to 110 percent at rated capacity, evidence of permanent deformation, inability to completely discharge the contents, binding of the gates, or any indication of leakage 1 minute after the gates are closed shall constitute failure of the test.

4.5.2.3 Grout slurry test. The bucket shall be loaded to 110 percent of rated capacity with grout slurry. The test specified in 4.5.2.2 shall be repeated.

4.6 Inspection of packaging. The packaging of the buckets shall be examined to determine compliance with the quality conformance inspection provisions of MIL-C-3580 and the requirements in section 5 herein.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Levels A and B. Preservation, packing, and marking for levels A and B shall be in accordance with the preservation, packing, and marking requirements for "dippers and buckets" as shown for levels A and B in MIL-C-3580.

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5.1.2 Commercial. Commercial preservation, packing, and marking shall be in accordance with ASTM D 3951. Additionally, each bucket shall be marked with the cube and gross weight.

6. NOTES

6.1 Intended use. This specification covers the hand-operated sizes of concrete buckets to be used for transporting and placing of concrete and grout.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size of bucket required (see 1.2) and specification part number (see 1.3).
c. When a first article is not required (see 3.2).
d. The time frame required for submission of first-produced buckets and number of models required (see 3.2).
e. When the Government will conduct any or all of the first-produced examination and tests. When the Government will conduct some but not all of the first-produced examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. Color required when other than as specified (see 3.10).
g. Level of preservation, packing, and marking required (see 5.1).

6.3 First article. When a first article inspection is required, the items should be first-produced concrete buckets. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

Attachments crane
Buckets, concrete
Construction materials handling, concrete

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Preparing activity:
Army - ME

Project 3815-0097

User activity:
Navy - MC

Review activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-3753E	Buckets, Concrete, Hand-Operated, 3/4- to 2-Cubic Yard Capacity

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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